

Proteomic Analysis of Colorectal Cancer vs Healthy Tissue (CPTAC)

Name: Ekramul islam

Table of Contents

1. Initial Setup and Aims	1
2. Preprocessing	2
2.1 Contaminant Removal.....	2
2.2 Sample Group Assignment	2
2.3 Missing Value Assessment.....	2
2.4 Filtering.....	3
2.5 Imputation	3
2.6 Normalisation	4
3. Statistical Analysis and Volcano Plot.....	5
4. Proteins of Interest	6
5. Troubleshooting and Iteration	7
6. Reflection	8

1. Initial Setup and Aims

The dataset originates from the Clinical Proteomic Tumor Analysis Consortium (CPTAC) and compares colorectal cancer with healthy colon tissue. Fifty-four samples were analysed: 21 Solid Tissue Normal and 33 Primary Tumour, processed across six TMT batches (01_CPTAC through

06_CPTAC). TMT_131 served as the internal reference channel during the database search and is absent from the output file.

My goal was to check whether the dataset looked reliable enough to compare the two tissue groups, then identify proteins with clear abundance differences between healthy and tumour tissue. I did the analysis in Python using pandas, numpy, scipy, seaborn, and matplotlib.

2. Preprocessing

2.1 Contaminant Removal

Before doing anything else with the data, I inspected the Accession column and found 69 entries tagged with the string "#CONTAM". These represent exogenous proteins introduced during sample preparation, most commonly keratins from skin contact and trypsin from the digestion step. Keeping these rows would distort intensity distributions and inflate the apparent protein list, so I removed them first, leaving 10,225 proteins.

```
df_clean = df[~df['Accession'].str.contains('#CONTAM', na=False)].copy()
```

2.2 Sample Group Assignment

The 54 data columns are labelled by batch and TMT channel (e.g. "01_CPTAC_TMT_126") without indicating condition directly. I parsed the sample information file to map each column to Normal or Tumour, confirming 21 Normal and 33 Tumour samples interleaved across all six batches, which prevents condition and batch from being fully confounded.

2.3 Missing Value Assessment

Proteomics data commonly contains missing values because lower-abundance proteins fall below the detection threshold. This makes the missingness Missing Not At Random (MNAR), since absence correlates with low abundance. On average, 2,026 values were missing per sample (roughly 20%). Some 6,366 proteins were detected in all 54 samples, while a long tail had very sparse detection. No sample stood out as an obvious technical failure.

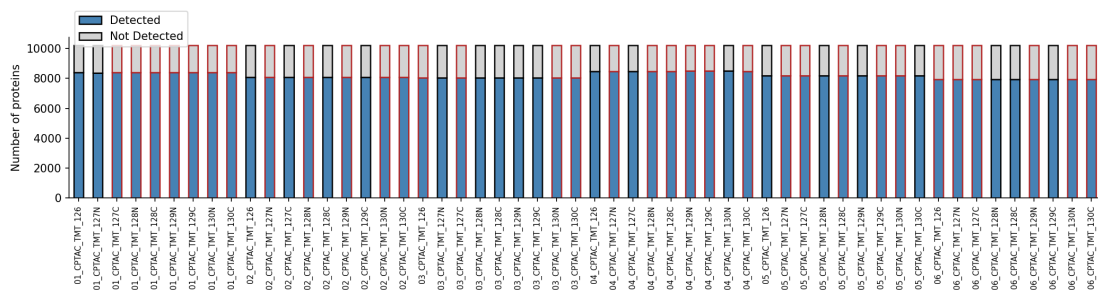


Figure 1. Proteins detected (blue) and missing (grey) per sample. Black edges indicate Normal samples; red edges indicate Tumour samples.

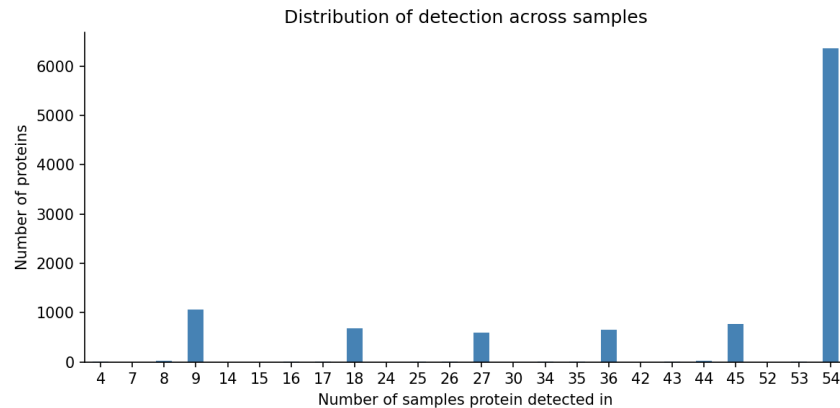


Figure 2. Distribution of detection frequency across all proteins. Many proteins are detected in all 54 samples, but a substantial tail is present in only a fraction of samples.

2.4 Filtering

I filtered out proteins missing in more than 75% of samples, on the basis that imputing the majority of values would make the downstream fold-change estimate more a product of the imputation model than real data. This removed 1,085 proteins, leaving 9,140. I considered a stricter 50% cutoff but decided against it, because proteins that are detected mainly in one condition and absent from the other could represent real biology rather than technical dropout, and I did not want to lose those. 75% is admittedly somewhat arbitrary and another analyst might draw the line differently.

```
missing_fraction = df_clean.iloc[:, 4:].isna().mean(axis=1)
df_filtered = df_clean[missing_fraction <= 0.75] # 9,140 proteins retained
```

2.5 Imputation

Missing values were filled using downshifted Gaussian imputation following the Perseus method. Because the missingness is MNAR, values absent near the detection limit are better replaced with draws from a shifted distribution than with zeros or column means. For each sample on the log10 scale, the imputation distribution was centred at the mean minus 1.8 SD with a width of 0.3 SD, then back-transformed. Zero missing values remained. The shift and width parameters came from the course material rather than being independently optimised for this tissue type.

```
for column in log10_data_cols:
    non_missing_values = log10_data_cols[column].dropna()
    std = non_missing_values.std()
    mean = non_missing_values.mean() - (1.8 * std)
    imputation_width = std * 0.3
    imputed_values = rng.normal(loc=mean, scale=imputation_width, size=n_missing)
```

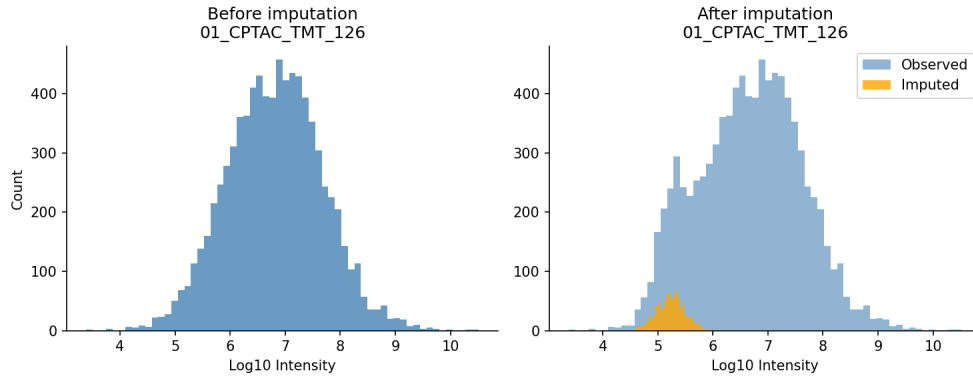


Figure 3. Log10 intensity distribution before (left) and after (right) imputation for one example sample. Orange bars show imputed values in the lower tail.

2.6 Normalisation

Even after within-batch TMT normalisation during the database search, systematic differences in total protein loading can persist. Per-sample medians before normalisation ranged from 2,560,533 to 8,717,021, a 3.4-fold difference that made normalisation clearly necessary. I applied median normalisation by dividing each sample by its median and rescaling to the global median. This assumes most proteins are unchanged, which is reasonable at this scale. Quantile normalisation was considered but not used because it forces identical distribution shapes, potentially removing real biological differences between groups. After normalisation all 54 sample medians converged to 5,836,685.

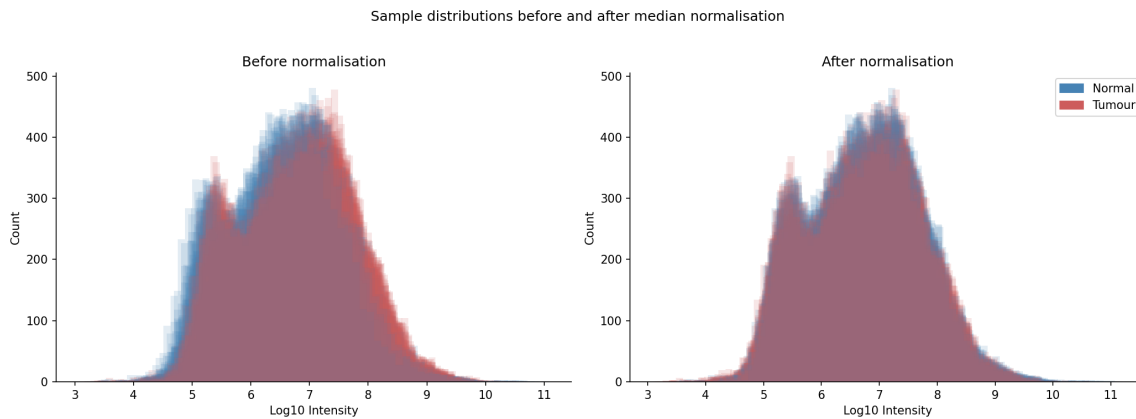


Figure 4. Log10 intensity distributions for all 54 samples before (left) and after (right) median normalisation. Blue = Normal, red = Tumour.

3. Statistical Analysis and Volcano Plot

For each of the 9,140 proteins, a Welch t-test compared the 21 Normal and 33 Tumour samples on log2-transformed intensities. Welch's variant was preferred over the equal-variance t-test because unequal group sizes and likely greater variance in cancer tissue make equal variance unlikely. Raw intensities were not used because proteomics data is approximately log-normally distributed, violating the normality assumption; log2 transformation helps address this. An empirical Bayes approach such as limma would improve sensitivity by sharing variance across proteins. Log2 fold change was the difference of group means on the log2 scale.

```
log2_normal = np.log2(normal_vals)
log2_tumour = np.log2(tumour_vals)
log2fc      = np.mean(log2_tumour) - np.mean(log2_normal)
_, p_val    = ttest_ind(log2_tumour, log2_normal, equal_var=False)
```

A protein was called significant if it met both thresholds simultaneously: $|\text{Log2FC}| \geq 1$ (at least 2-fold change) and $-\log_{10}(\text{p-value}) \geq 1.3$ ($p < 0.05$). Requiring both prevents flagging proteins with a tiny effect size just because the p-value is small, or conversely proteins that show a large fold change based on noisy measurements. Of the 9,140 proteins tested, 83 were significantly upregulated in tumour, 506 were significantly downregulated, and 8,551 showed no significant change. I also checked the list with Benjamini-Hochberg correction: 544 proteins still passed $q \leq 0.05$ with the same absolute fold-change threshold. The larger number of downregulated proteins is consistent with a broad loss of normal colon tissue identity in the tumour proteome.

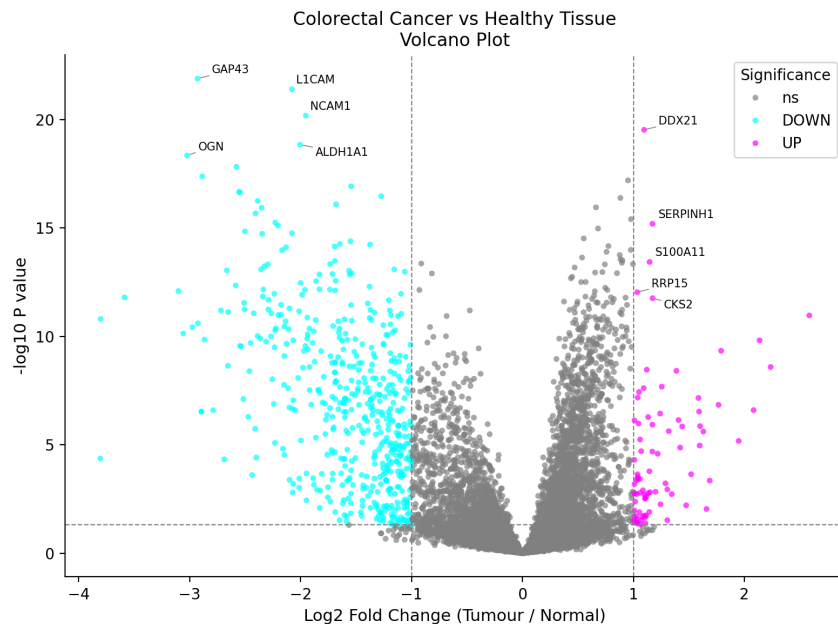


Figure 5. Volcano plot of all 9,140 tested proteins. Magenta = significantly UP ($n=83$), cyan = significantly DOWN ($n=506$), grey = not significant. Dashed lines mark $|\text{Log2FC}|=1$ and $-\log_{10}P=1.3$. Top hits are labelled by gene symbol.

4. Proteins of Interest

A shortlist of ten proteins was selected on the basis of highest statistical confidence: the top five upregulated and top five downregulated by $-\log_{10}(\text{p-value})$, summarised in Table 1.

Gene	Log2FC	$-\log_{10}P$	Direction
DDX21	+1.10	19.5	UP
SERPINH1	+1.17	15.2	UP
S100A11	+1.14	13.4	UP
RRP15	+1.03	12.0	UP
CKS2	+1.17	11.8	UP
GAP43	-2.93	21.9	DOWN
L1CAM	-2.08	21.4	DOWN
NCAM1	-1.95	20.2	DOWN
ALDH1A1	-2.00	18.8	DOWN
OGN	-3.02	18.3	DOWN

Table 1. Shortlist of ten proteins of interest. Log2FC and $-\log_{10}P$ values are from the Welch t-test on $\log_2$ -transformed normalised intensities.

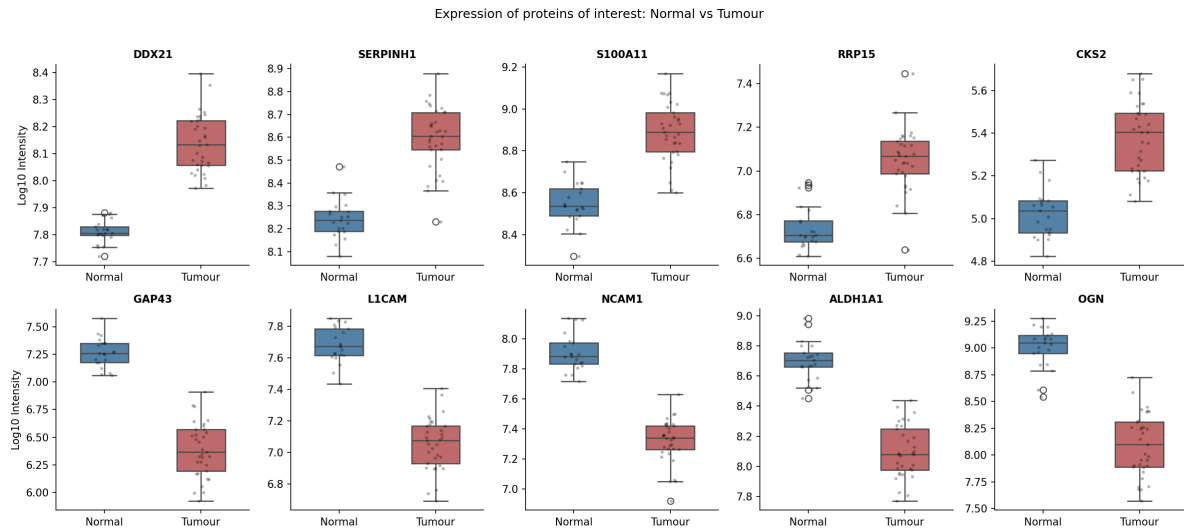


Figure 6. $\log_{10}$ intensity distributions for the ten shortlist proteins in Normal (blue) and Tumour (red) samples. Points show individual sample values.

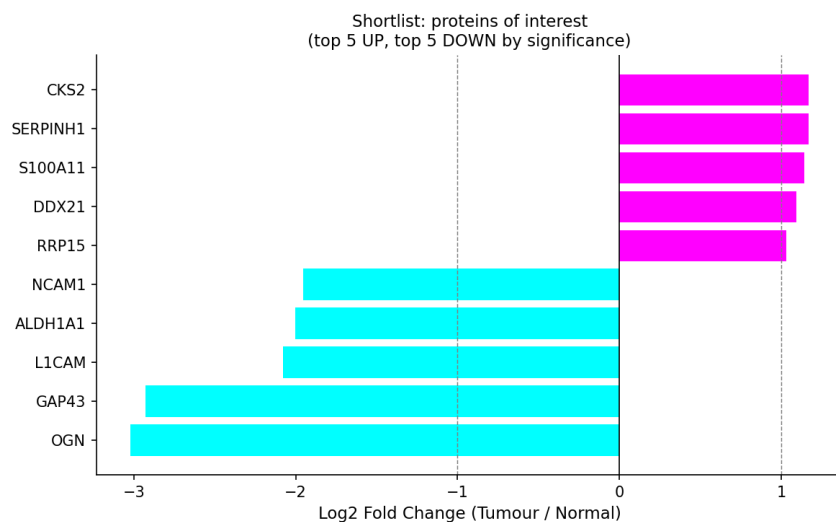


Figure 7. Log2 fold change for shortlist proteins. Magenta = upregulated in tumour; cyan = downregulated.

The upregulated proteins are consistent with active proliferation. DDX21 and RRP15 support ribosome biogenesis, reflecting elevated translational demand in dividing tumour cells. SERPINH1, a collagen-specific chaperone, points to extracellular matrix remodelling. S100A11 is a calcium-binding S100 protein with documented roles in CRC invasion. CKS2 is a cyclin-dependent kinase regulator frequently elevated in CRC. The downregulated proteins reflect a loss of normal colon tissue identity. GAP43, L1CAM, and NCAM1 are associated with neuronal and normal mucosal cell types; their loss is consistent with dedifferentiation in colorectal cancer. ALDH1A1 (Log2FC -2.00), an aldehyde dehydrogenase involved in retinoic acid biosynthesis, points to disrupted retinoid signalling, which may reflect altered epithelial differentiation. OGN (mimecan) is a stromal matrix protein whose reduction reflects tumour microenvironment remodelling.

5. Troubleshooting and Iteration

Parsing the sample information file took the most time: cells combined patient IDs and condition labels with embedded newlines that needed cleaning before string matching worked. The absence of TMT_131 was initially unexpected; the assessment sheet confirmed it as an internal reference channel.

The biggest correction came when I noticed the t-test had been run on raw intensities rather than log2 values. Switching to log2 made the comparison more defensible and matched how Perseus handles this step. For DDX21, the p-value shifted from 8.9e-14 to 3.0e-20, though the top-ranked proteins remained consistent in direction and interpretation. Gene labels on the volcano plot were initially overlapping, resolved using the adjustText library.

6. Reflection

Working through this dataset reinforced how much analytical work happens before the statistics. The filtering threshold and imputation strategy together determined which proteins could be tested. Dropping all rows with any missing values would have reduced the dataset to 6,366 proteins instead of 9,140, biasing results toward the most abundant. I ran BH correction as a check on the uncorrected p-value calls; finding that 544 of the 589 hits still passed $q \leq 0.05$ was reassuring. I also realised a paired t-test would be more appropriate for patients who contributed both tissue types, which I would address if repeating the analysis.

With more time I would run pathway enrichment on all 589 regulated proteins rather than interpreting only the top ten. The shortlist gives a useful starting point but does not tell you whether the changes cluster into particular biological processes, which would give a more complete picture of what is actually disrupted in the cancer tissue.